Practical 5

Aim: To create a **calculator application using a SOAP web service** to perform basic arithmetic operations and test it by passing proper inputs.

Operations:

- Addition
- Subtraction
- Multiplication
- Division

1️⃣ Create Web Application

Open **NetBeans**

Go to: File → New Project → Java Web → Web Application

Project Name: CalculatorWS

Server: GlassFish Server

Click **Finish**

2️⃣ Create SOAP Web Service

Right click **Project**

New → Web Service

Fill: Web Service Name: Calculator

Package: calc

Click **Finish**

3️⃣ Calculator Web Service Code

Open

Calculator.java

Replace code with this:

```
package calc;
import javax.jws.WebService;
import javax.jws.WebMethod;
@WebService(serviceName = "Calculator")
public class Calculator {
  @WebMethod
```

```
    public int add(int a, int b)
    {
        return a + b;
    }
    @WebMethod
    public int subtract(int a, int b)
    {
        return a - b;
    }
    @WebMethod
    public int multiply(int a, int b)
    {
        return a * b;
    }
    @WebMethod
    public double divide(int a, int b)
    {
        return (double)a / b;
    }
}
```

Save the file.

4️⃣ Run the Project

Right click project -> Run

Check WSDL in browser: http://localhost:8080/CalculatorWS/Calculator?wsdl

If XML page appears → service is working ✅

5️⃣ Create Web Service Client

Right click project

New → Web Service Client

Enter WSDL URL: http://localhost:8080/CalculatorWS/Calculator?wsdl

Click **Finish**

NetBeans will generate client classes automatically.

6️⃣ Create JSP Client

Create file: calculator.jsp

7️⃣ JSP Code

Paste this code.

```
<%@page contentType="text/html"%>
<html>
<body>
<h2>Calculator using SOAP Web Service</h2>
<%
try
{
  calc.Calculator_Service service = new calc.Calculator_Service();
  calc.Calculator port = service.getCalculatorPort();
  int a = 20;
  int b = 10;
  out.println("Addition = " + port.add(a,b));
  out.println("<br>");
  out.println("Subtraction = " + port.subtract(a,b));
  out.println("<br>");
  out.println("Multiplication = " + port.multiply(a,b));
  out.println("<br>");
  out.println("Division = " + port.divide(a,b));
}
catch(Exception e)
{
  out.println(e);
}
%>
</body>
</html>
```

8️⃣ Run JSP

Open browser: http://localhost:8080/CalculatorWS/calculator.jsp

Example Output:

Addition = 30

Subtraction = 10

Multiplication = 200

Division = 2.0

Libraries Used

Built-in Java libraries:

javax.jws.*

JAX-WS

GlassFish SOAP Engine

No external libraries required.